

Generalised Linear Models

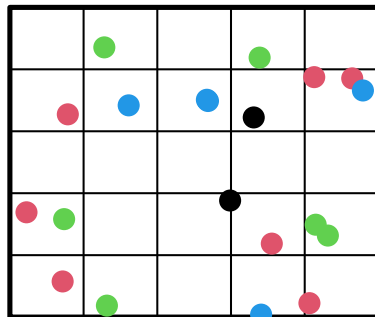
for data of multiple species

Bert van der Veen

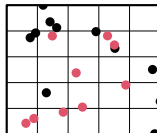
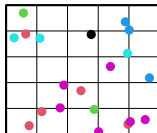
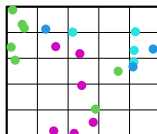
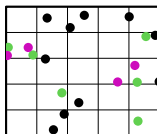
University of Bayreuth

A meadow in the Dutch dunes.

How many plant species are there in this meadow?



We go into the meadow and count plants in quadrats. We find 4 species.



We resample the dune meadow, and find different numbers of species: 3, 3, 6, 2. And different compositions. On average we have found 3.5 species per quadrat.

Sampling variation



If we sample many times, we have an accurate picture of the whole meadow (and variability in the number of species in a quadrat that we might find).

Statistical modeling

Instead of a focus on data, we consider the *data generating process*

- ▶ We collect data
- ▶ Decide on a research question for *the population*
- ▶ Learn about the variation in the data
 - ▶ Which requires formulating a model
- ▶ Work out distribution of the estimates
 - ▶ And find the “best” estimate
- ▶ Conclude if our answer is robust for the population (e.g., fields like this have more than 6 species)

Generalised linear models (GLMs)

GLMs as a framework were introduced by Nelder and Wedderburn (1972) uniting many different models. With a special focus on teaching statistics.

- ▶ Linear regression
- ▶ Logistic regression
- ▶ Probit regression
- ▶ Complementary log-log regression
- ▶ Log-linear regression
- ▶ Gamma regression

Generalised Linear Models

For when the assumptions of linear regression fail; GLMs extend the framework to address bounded responses and mean-dependent variance.

Assumptions of linear regression:

- ▶ Linearity (straight line)
- ▶ Independence of errors
- ▶ Homoscedasticity (same variance for all errors)
- ▶ Normality (distribution of errors)

Generalised Linear Models

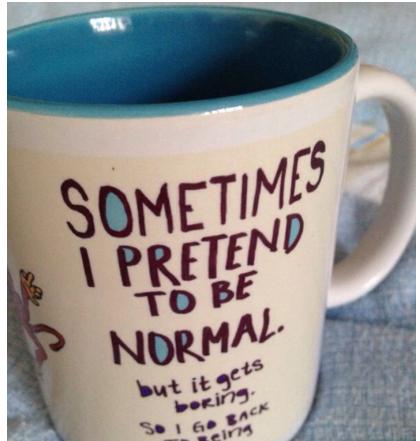
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GLMs address:

- ▶ Variance changes with the mean



Components of a GLM

- ▶ Systematic component: η
- ▶ Random component:
data/distribution
- ▶ The link function: connects
these components (not a
data transformation)
- ▶ The variance function

**No explicit error term;
parameters estimated by MLE
from the exponential family
likelihood:**

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likelihood:**

$$\mathcal{L}(y_i; \Theta) = \exp \left\{ \frac{y_i \eta_i - b(\eta_i)}{a(\phi)} + c(y_i, \phi) \right\} \quad (1)$$

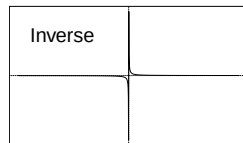
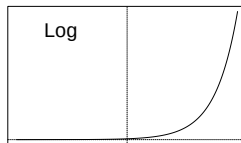
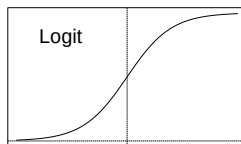
Generalised linear model

$$\begin{aligned}g\{\mathbb{E}(y_i|x_i)\} &= \eta_i = \alpha + x_i\beta \\ \mathbb{E}(y_i|x_i) &= g^{-1}(\eta_i) = g^{-1}(\alpha + x_i\beta)\end{aligned}\tag{2}$$

$g(\cdot)$ is the **link function**

The link function

- ▶ Is a smooth/monotone function
- ▶ Has an inverse $g^{-1}(\cdot)$
- ▶ Restricts the scale
- ▶ $g(\cdot)$ can be e.g.



Variance function

Variance changes with the mean:

$$\text{var}(y_i; \mu_i, \phi) = \frac{\partial^2 b(\eta_i)}{\partial \eta_i^2} a(\phi)$$

- ▶ ϕ : the dispersion parameter, constant over observations
 - ▶ Fixed for some response distributions
- ▶ $a(\phi)$ is a function of the form ϕ/w_i (McCullagh and Nelder 1989)

Often used distributions in ecology

- ▶ Binomial: occurrence/counts. Presence of species, number of germinated seeds out of a total
- ▶ Poisson: counts. Abundance
- ▶ Negative binomial (fixed dispersion): counts. Number of species or abundance
- ▶ Gamma: (positive) continuous. Body size or biomass
- ▶ Ordinal (cumulative link). Cover classes
- ▶ Beta (logit link). Cover (note: not a GLM)

The binomial GLM

$$p(y_{ij} = 1) = p_{ij} = g^{-1}(\eta_{ij}) \quad (3)$$

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Link functions:

- ▶ Logit: $\log\left(\frac{\pi_i}{1-\pi_i}\right)$ and inverse $\frac{\exp(\eta_i)}{1+\exp(\eta_i)}$ - *the canonical link*
- ▶ Probit: $\Phi^{-1}(\pi_i)$ and inverse $\Phi(\eta_i)$
- ▶ Complementary log-log: $\log(-\log(1 - \pi_i))$ and inverse $1 - \exp(-\exp(\eta_i))$
- ▶ ~~Log-log~~
- ▶ Logit is canonical and easier to interpret
- ▶ Probit is sometimes easier mathematically than Logit
- ▶ Complementary log-log for counts

The Poisson and negative binomial GLM

For count responses, the canonical link is log:

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- ▶ Negative binomial: $\text{var}(y_{ij}) = \mu_{ij} + \mu_{ij}^2/\phi$ (overdispersion)

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Count data in ecology is almost always overdispersed; the negative binomial is the default choice.

Example: Swiss bird occurrence

Observation process: see if a bird is present (or we might hear it)

Alternatively: The proportion of a species in a place

Alternatively: Count of birds in the forest

There are often many ways to observe the same ecological process.
We need to **disentangle** this from the ecological process.

Example: Swiss birds

- ▶ Data by Schmid et al. (1998): the Swiss breeding bird atlas
- ▶ Occurrence of 56 species at 2524 locations recorded over a 4-year period

Gartenrotschwanz

Phoenicurus phoenicurus
Bourgonne à front blanc
Codonomus caninus
cauchischia d'art
Common Redstart

Rote Liste: potenziell gefährdet (NT)
Bestand: 12'000-18'000 Paare (2013-2016)

Hotspot: Basel-Stadt, Basel-Landschaft, Basel-Stadt, Basel-Landschaft, Basel-Stadt, Basel-Landschaft



Der Gartenrotschwanz kommt in der ganzen Schweiz vor, oft aber nur in geringer Dichte. Am häufigsten ist er im Süden und in den grossen Tälern des Tessins, im Mosaik-Gebiet und im Bergell GR, im Mittelland und Oberrhein sowie in der Region Basel. Rund 90% des Bestands leben unter 1500 m. Zwei Brücken gewählt werden Orte, wo gleichzeitig Bläue, nacheinander Boden und Höhen vorhanden sind. Die kleinen eher ländliche Wohnquartiere, Parks, Familiengärten, Rebberge mit Bläuen, Hochstamm-Obstgärten oder kleine Wälder sein. Seltener werden bis auf 2500m bei Tscham VS und in den Jahren 2013-2016, bis auf 2200m bei S-charf GR (M. Brändt beobachtet). Der höchste Brutnachweis stammt von 2200m bei Zermatt (CH). In La Chaux-de-Fonds NE hat man auf 1 km² 31 Reviere gezählt¹. In Basel BS lag die maximale Dichte bei 6,1 Reviere/1 ha². Für den Schweizer Bestand waren die Daten der Sechzigjahre in der Subalpine verschoben³. Der Rückgang tritt nicht mehr in der Neuzugangsphase fort. Erst seit 2002 zeigt sich eine leichte Erholung. Regional unterscheiden sich die Trends jedoch stark. In den ländlichen Gebieten nördlich der Alpen dominieren die bereits geringen Bestände weiter aus. Die Verluste betragen im Kanton

Zürich von 1988 bis 2008 85%⁴, am Bodensee von 1980 bis 2010 rund 90%⁵ und im Kanton Basellandschaft von 1983 bis 2013 42%⁶. Den starken Einbruch in Höhen von 300 bis 500m, die eine Folge des Verschwindens von Hochstamm-Obstgärten, intensiveren Grünlandbewirtschaftung und des Fehlens offener Bodenstellen für die Jungvögel und⁷, stellt ein geringer Bestand dar, eine zusätzliche Gefahr darstellen⁸. Im Wallis und im Tessin nehmen die Bestände selbst in tiefen Lagen zu⁹ und tragen damit ebenfalls zum derzeit positiven bestandsweiten Trend bei. Die maximal 97 Reviere auf der 3,1 km² grossen Waldbrände bei Leuk VS¹⁰ zeigen das enorme Beseitigungspotential dieser Art. In Italien und Frankreich sind die Trends stark positiv^{11,12}. In Deutschland und Österreich sind sie trotz regionaler Einbrüche¹³ ausserordentlich positiv¹⁴. Die europäische Bestandsliste von 1980 bis 2014 leicht positiv¹⁵.

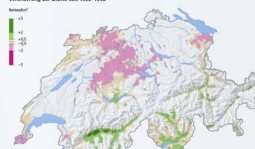
Jürgens Lesau, Nicolas Martinet & Boris Dettl

Gartenrotschwanz

Dichte 2013-2016



Veränderung der Dichte seit 1993-1996



Kartenwerk
Andreas Bopp, Lucien
Gardner, Jörg
Gerrit Dagg, Hartmut

The data

Falco_subbuteo	Anthus_trivialis	Phylloscopus_bonelli	Tetrao_tetrix	Parus_caeruleus	Dendrocopos_major
0	1	0	1	0	1
1	0	0	0	1	1
0	0	0	0	1	1
0	1	0	0	1	1
0	0	1	1	0	0
1	0	0	0	1	1

The environmental variables

avg	cov	cv	dns	fhd	p10	p25	p95	rt_p2595	std	uhd
13.549999	34.1	0.6287823	34.1	1.8422778	2.68	5.950000	28.14	0.2114428	8.52	2.381401
13.790000	32.0	0.5547498	32.0	1.8031981	3.96	7.090000	28.20	0.2514184	7.65	2.276368
19.340000	14.3	0.5351603	14.3	2.0021141	4.79	9.139999	34.03	0.2685865	10.35	2.364149
15.460000	34.5	0.4618370	34.5	1.7484871	5.19	9.790000	26.36	0.3713961	7.14	2.369797
2.290000	2.0	0.7292576	2.0	0.3038808	1.06	1.200000	5.74	0.2090592	1.67	1.270930
8.929999	61.3	0.5890258	61.3	1.4213525	2.21	4.570000	18.05	0.2531856	5.26	2.392563

- ▶ Bioclimatic variables (bioclim)
- ▶ Topography (slope, aspect, TPI, TWI) from a DEM
- ▶ Potential evapotranspiration (PET) from solar radiation
- ▶ Moisture index, degree days above zero
- ▶ Vegetation structure from LiDAR

Data format

There are two ways to format these data:

Wide format: Species as columns (as presented)

Long format: Species is one column, and “Site” is another column

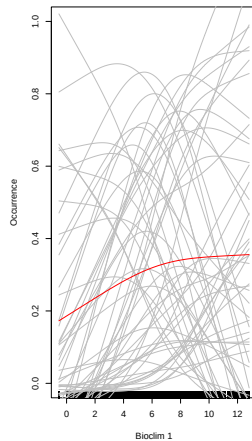
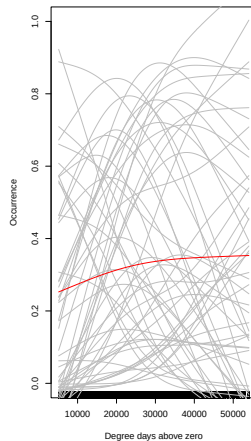
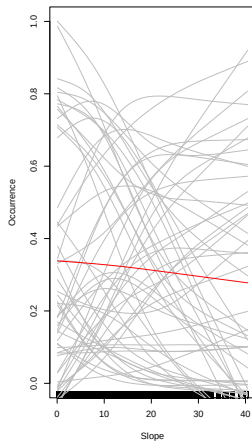
The format of the data does not affect the model. Some functions accept long format, other wide format, but the formulation of the model is up to us.

Swiss birds: to long format

```
data <- data.frame(y, X)
datalong <- reshape(data,
                     varying = colnames(y),
                     v.names = "occ",
                     idvar = "Site",
                     timevar = "Species",
                     direction = "long")

datalong$Species <- factor(datalong$Species,
                           labels = colnames(y))
```

Swiss birds: visually inspect the data



Is there a community-level trend?

Swiss birds: fit a model

```
model1 <- gllvm::glmmVA(occ ~ slp,  
                        data = datalong, family = "binomial")  
coef(model1, "row.params.fixed")
```

```
##           slp  
## -0.004001257
```

Here we assume that the intercept and slp effect are the same for all species

Multispecies modeling

- 1) Is the same effect for all species realistic?
- 2) Is the same (average) probability of occurrence for all species realistic?

Multispecies modeling

- 1) Is the same effect for all species realistic?
- 2) Is the same (average) probability of occurrence for all species realistic?
- 3) We usually assume that species have their own preferred environmental conditions
- 4) Some species might still like similar conditions; there is a common component
- 5) We can separate this out with GLMMs or with a “sum-to-zero” contrast

Swiss birds: species-specific effects

```
model2 <- glvm::glmmVA(occ ~ slp*Species,
                        data = datalong, family = binomial)
```

- ▶ One intercept per species
- ▶ One slp effect per species
- ▶ But all are relative to the first species

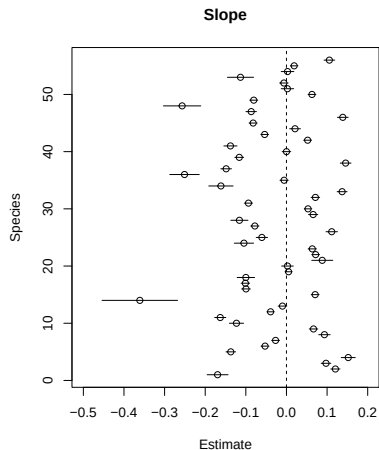
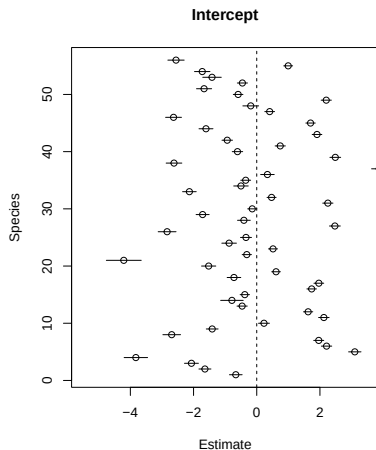
Swiss birds: slopes without reference

```
model3 <- gllvm::glmmVA(occ ~ Species + slp:Species,
                        data = datalong, family = binomial)
```

The same model, but a bit easier to interpret

- ▶ One intercept per species
- ▶ One slp effect per species
- ▶ Slopes are not relative to each other (prevents post-hoc processing of tests and CI)

Swiss birds: results



Interpreting Binomial GLM coefficients

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$$0.52 \cdot \exp(-0.17) = 0.52 \cdot 0.84 = 0.43$$

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Contrasts

There are other “contrast” treatments in R than “dummy”

- ▶ We can instead use “sum-to-zero” contrasts
 - ▶ If the sum is zero, the mean must be too
 - ▶ The coefficient of the last species is set to the negative sum

```
contrasts(datalong$Species) <- contr.sum(levels(datalong$Species))
contr.sum(4)
```

```
##      [,1] [,2] [,3]
## 1      1      0      0
## 2      0      1      0
## 3      0      0      1
## 4     -1     -1     -1
```

Swiss birds: species-specific responses with common effect

```
model4 <- glvm::glmmVA(occ ~ slp + Species + slp:Species, data = datalong,
                        family = binomial)
```

```
##           slp
## -0.02703652
```

- ▶ One intercept per species
- ▶ One slp effect that is the same for all species (the mean of effects)
- ▶ One slp effect per species, relative to the common effect

The benefit: the average effect gets a statistical test.

By design corresponds to the result from our previous model:

```
mean(coef(model3, "row.params.fixed")[grep("slp",
names(coef(model3, "row.params.fixed")))] = -0.0270369
```

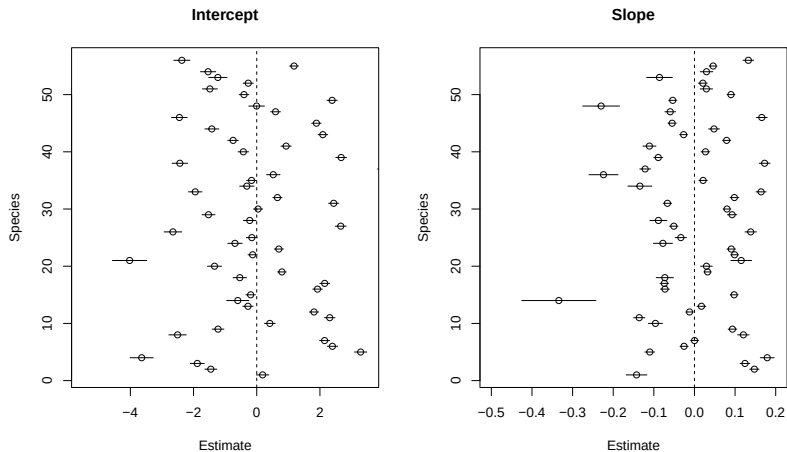
Swiss birds

The three models have the same number of parameters, but are just differently parameterized. So, their log-likelihoods and AIC are the same:

```
AIC(model2, model3, model4)
```

```
##           df          AIC
## model2 112 132240.8
## model3 112 132240.8
## model4 112 132240.8
```

Swiss birds: species-specific deviations to common effects



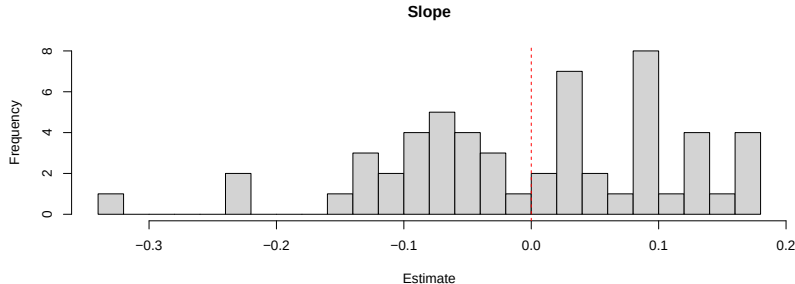
Average intercept: -0.848, 0. Average slope: -0.027, 0.

Swiss birds: conclusions

We can conclude that fewer bird species occur in steep places

Most species are more negatively affected than the average

Some species are positively affected by slope, but most negatively



Interpreting the coefficients

Or with predict:

```
predict(model3, newX =
  data.frame(Species = factor("Falco_subbuteo", levels = colnames(y)), slp = 1),
  type = "response")
```

```
##           [,1]
## 1 0.3031189
```

The ecological process

What do we know of the processes that generate these data?

- ▶ Meta-community theory
- ▶ Assembly processes (filtering)
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Multispecies models provide a statistical connection to these ecological frameworks. We do not just use a fancy tool, we use a fancy tool because we believe it aligns well with our understanding of the ecological process.

The ecological process (2)

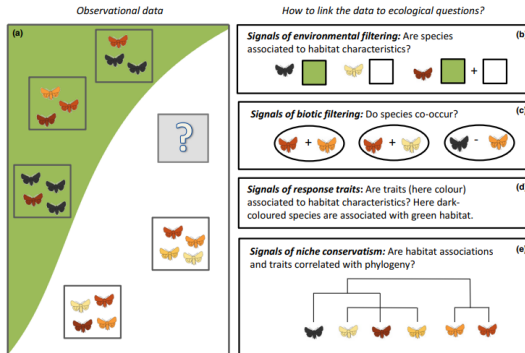


Figure 2 A conceptual illustration of some key questions in community ecology. The green and white colours represent differences in the environmental

Figure 3: Figure 2 from Ovaskainen et al. (2017)

On ecological communities

The concept of an ecological community is of limited use. By definition:

An ecological community is a group or association of two or more species occupying the same geographical area at the same time

- ▶ We often think of ecological communities as groups
- ▶ We can also think of a community as a continuum that changes along a gradient (Austin 1985)
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- ▶ We can also think of them as the species in our data

In the models we fit: each species gets its own intercept (baseline occurrence or abundance) and its own slope (response to the gradient). The community is the collection of those species-level curves, estimated jointly.

An ecological gradient

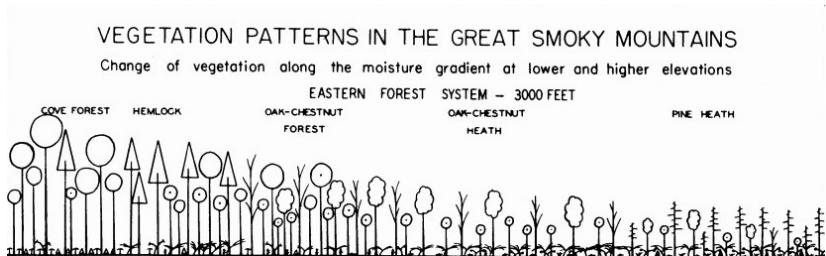


Figure 4: Whittaker (1956)

Response curves

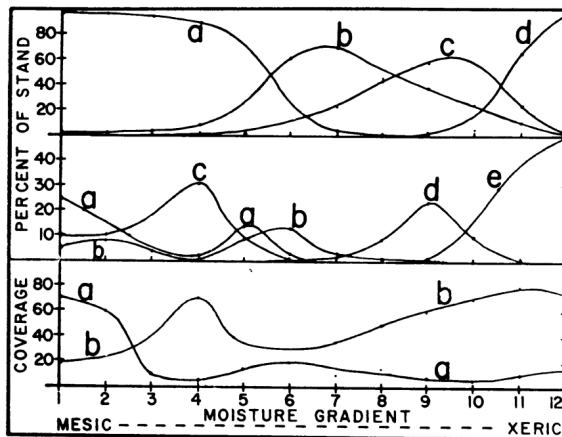


FIG. 4. Transect of the moisture gradient, 3500-4500 ft. Top—curves for tree classes; a, mesic; b, submesic; c, subxeric; d, xeric. Note expansion of mesic stands, compared with Figs. 2 and 3. Middle—curves for tree species: a, *Tilia heterophylla*; b, *Halesia monticola* (both

Estimating response curves

Whittaker's curves show each species has its own optimum and tolerance along an environmental gradient. In a statistical model:

- ▶ The intercept captures the species' baseline abundance
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- ▶ Together: one response curve per species

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When data are too sparse to estimate one curve per species, ordination is the tool of choice: that is Day 3.

The common effect is the Whittakerian community signal; species-specific deviations are the Gleasonian individual responses

Example: macroinvertebrate counts in USA desert

- ▶ Data by Pina and Loughheed 2022
- ▶ Counts of 13 macroinvertebrate species in 14 wetlands
- ▶ NO3 concentration as an ecological gradient: species sort along it via environmental filtering

Observation process: count of macroinvertebrates in three “dips”

Alternatively: proportion of a species in a dip

Alternatively: presence/absence



Figure 6: Daniel J. Calderon
slp.usace.army.mil

The abundance data

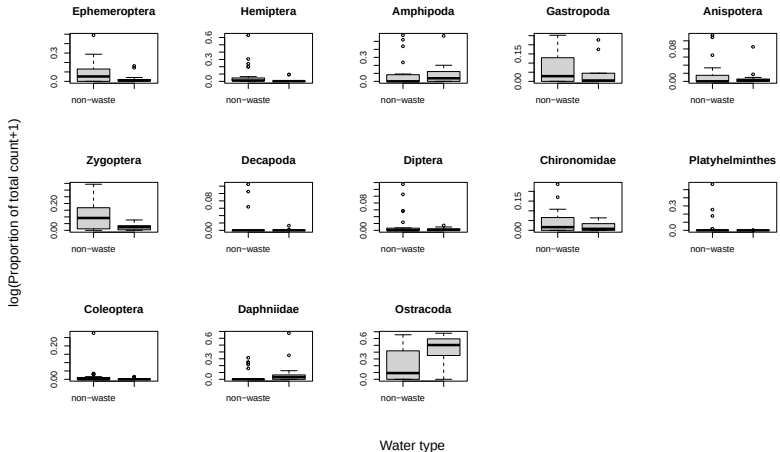
Ephemeroptera	Hemiptera	Amphipoda	Gastropoda	Anisoptera	Zygoptera	Decapoda	Diptera	Chiron
0	1	100	8	0	1	12	0	
21	0	0	5	3	10	0	1	
11	3	5	0	4	32	0	0	
0	1	11	0	0	0	0	0	
80	1	60	25	0	7	0	2	
9	15	0	6	0	10	0	1	
30	0	0	10	7	25	0	1	
10	1	60	190	4	60	0	0	
6	0	0	5	0	3	2	2	
5	41	0	5	0	2	0	0	
32	2	0	0	1	35	0	0	
6	25	0	31	2	13	0	0	
28	15	400	0	21	14	0	0	
28	15	400	0	21	14	0	0	
6	150	200	70	2	48	3	4	
6	150	200	70	2	48	3	4	
9	1	0	35	14	50	0	7	
26	1	0	1	10	21	0	12	
0	1	0	2	2	7	0	0	
1	0	60	6	1	9	0	0	
13	0	15	26	0	8	0	1	
87	4	0	0	0	3	0	0	
2	1	11	4	2	1	0	1	

The environment data

Year	Hydro	Water_Type	Conductivity	DOC	TDN	Turbidity	Alkalinity	Total_CHL	Correc
2018	permanent	non-waste	4.060	2.846	0.306	4.40	63.050	2.231	
2018	permanent	waste	2.582	23.160	3.544	60.73	412.400	95.211	
2018	permanent	non-waste	8.563	28.120	2.450	17.40	363.708	16.915	
2018	permanent	non-waste	15.710	75.040	7.160	38.50	457.625	26.160	
2018	ephemeral	non-waste	1.029	4.012	0.386	3.78	198.042	12.657	
2018	ephemeral	non-waste	1.204	5.356	0.491	24.70	168.042	30.353	

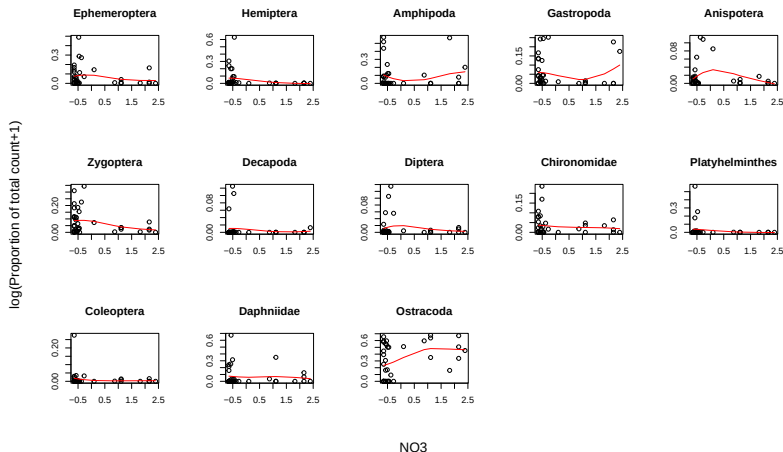
- ▶ 11 environmental variables
 - ▶ Water chemistry
 - ▶ Water type
 - ▶ Presence of hydro power
 - ▶ Permanent or temporary wetland

Visually inspect the data: categorical covariate



Is there a common effect?

Visually inspect the data: continuous covariate



Is there a common effect?

Wetlands: species-specific responses with common effect

```
model5 <- glvm::glmmVA(Count ~ Species + N03 + N03:Species, data = long,
  family = "poisson")
```

##	Species1	Species2	Species3	Species4	Species5
##	0.46550197	0.15521206	1.65736757	0.34595669	-0.99528664
##	Species6	Species7	Species8	Species9	Species10
##	0.49883688	-2.72196321	-1.75702484	-0.02330576	-1.26763733
##	Species11	Species12	N03	Species1:N03	Species2:N03
##	-1.73786959	1.81756745	-0.02504060	0.04674952	-0.74506201
##	Species3:N03	Species4:N03	Species5:N03	Species6:N03	Species7:N03
##	0.21656738	0.51506112	0.14243956	0.12796826	-0.43764445
##	Species8:N03	Species9:N03	Species10:N03	Species11:N03	Species12:N03
##	0.06845048	0.34248926	-0.95659236	0.48870132	-0.05675743

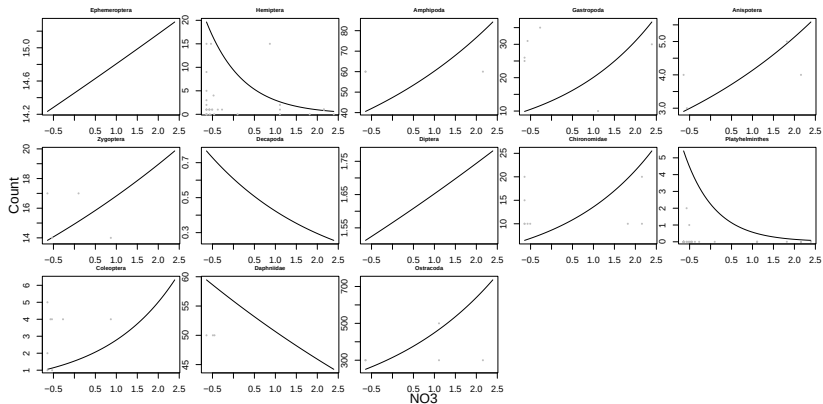
Wetlands: species-specific responses with common effect

Count data is usually overdispersed; we might want to switch to a NB.

```
model6 <- glvm::glmmVA(Count ~ Species + N03 + N03:Species, data = long,
  family = "negative.binomial")
```

##	Species1	Species2	Species3	Species4	Species5
##	0.48865688	0.06039061	1.67555354	0.38647231	-0.97793317
##	Species6	Species7	Species8	Species9	Species10
##	0.52117783	-2.68022268	-1.73405743	-0.02496534	-1.37336482
##	Species11	Species12	N03	Species1:N03	Species2:N03
##	-1.75168084	1.84013032	-0.03750368	0.06151000	-1.10547955
##	Species3:N03	Species4:N03	Species5:N03	Species6:N03	Species7:N03
##	0.27812231	0.46946945	0.25306544	0.15671279	-0.32427073
##	Species8:N03	Species9:N03	Species10:N03	Species11:N03	Species12:N03
##	0.09137534	0.48958046	-1.33012049	0.62805941	-0.06027213

Wetlands: response curves



Wetlands: conclusions

Species sort along the NO_3 gradient consistent with environmental filtering: some tolerate high NO_3 , most are negatively affected.

The average effect of NO_3 is negative (but not statistically significant at this sample size)

Species-specific responses are heterogeneous: some are positively affected, most negatively

Interpreting the coefficients

```
##      Species1      Species2      Species3      Species4      Species5
##  0.48865688    0.06039061    1.67555354    0.38647231   -0.97793317
##      Species6      Species7      Species8      Species9      Species10
##  0.52117783   -2.68022268   -1.73405743   -0.02496534   -1.37336482
##      Species11     Species12      N03    Species1:N03    Species2:N03
## -1.75168084    1.84013032   -0.03750368    0.06151000   -1.10547955
## Species3:N03 Species4:N03 Species5:N03 Species6:N03 Species7:N03
##  0.27812231    0.46946945    0.25306544    0.15671279   -0.32427073
## Species8:N03 Species9:N03 Species10:N03 Species11:N03 Species12:N03
##  0.09137534    0.48958046   -1.33012049    0.62805941   -0.06027213
```

- ▶ Negative means a decrease in the response and positive increase
- ▶ More specifically here: the coefficient is multiplicative decrease in $\exp(\text{intercept})$ for a unit change in N03
- ▶ E.g., for “Ephemeroptera”: $\exp(2.671) \cdot \exp(-0.038 + 0.062) = 14.46 \cdot 1.024$

What NO₃ does not explain

Species responses to NO₃ account for part of the community structure. But species also co-occur in structured ways that NO₃ alone does not capture:

- ▶ Biotic interactions
- ▶ Unmeasured environmental gradients
- ▶ Shared evolutionary history

What NO₃ does not explain

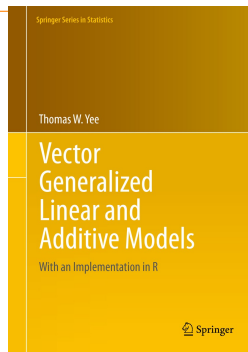
Species responses to NO₃ account for part of the community structure. But species also co-occur in structured ways that NO₃ alone does not capture:

- ▶ Biotic interactions
- ▶ Unmeasured environmental gradients
- ▶ Shared evolutionary history

Accounting for this residual co-occurrence structure requires modelling species associations directly. That is the topic of Days 2 and 3.

Vector GLMs

- ▶ One GLM per species
- ▶ Each species gets its own dispersion parameter ϕ_j (vs. a shared ϕ)
- ▶ Slightly more flexible than what we have done so far



Methods in Ecology and Evolution



Methods in Ecology and Evolution 2012, **3**, 471–474

doi: 10.1111/j.2041-210X.2012.00190.x

mvabund – an R package for model-based analysis of multivariate abundance data

Yi Wang^{1,2}, Ulrike Naumann¹, Stephen T. Wright¹, and David I. Warton^{1,3*}

¹School of Mathematics and Statistics, The University of New South Wales, Sydney, NSW 2052, Australia; ²School of

Fitting vector GLMs

A few software implementations exist:

- ▶ The VGAM R-package
- ▶ The glmmTMB R-package
- ▶ The gl1vm R-package

Clearly, we will use the last one.

gllvm

Received: 7 May 2019

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DOI: 10.1111/2041-210X.13303

APPLICATION

Methods in Ecology and Evolution



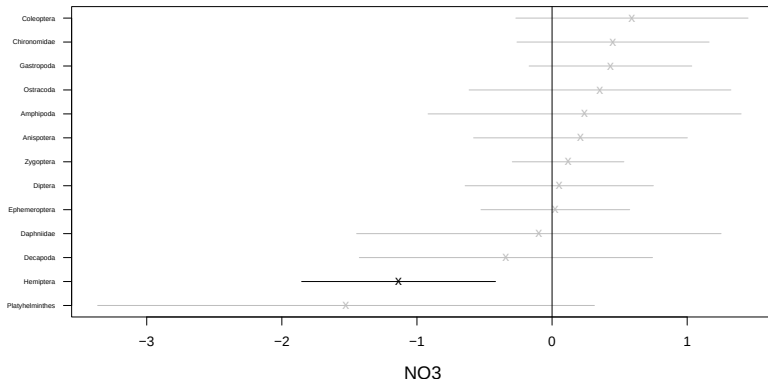
gllvm: Fast analysis of multivariate abundance data with generalized linear latent variable models in R

Jenni Niku¹ | Francis K. C. Hui² | Sara Taskinen¹ | David I. Warton³

- ▶ Originally published in 2019 by Niku et al. I “joined in” shortly after
- ▶ For model-based multivariate analysis of community ecological data
- ▶ Models are fitted in C++ (Kristensen et al. 2015)
- ▶ Can fit many different models: VGLM(M), JSDM, and ordination

VGLM in gllvm

```
model7 <- gllvm::gllvm(y, X = X, formula = ~N03, family = "negative.binomial", num.lv = 0)
gllvm::coefplot(model7)
```



(the dummy contrasting is now implicit)

Does NO3 improve the model?

```
model8 <- gllvm::gllvm(y, formula = ~1, family = "negative.binomial", num.lv = 0)
anova(model7, model8)
```

```
## Model 1 : y ~ NULL
## Model 2 : ~ NO3
```

```
##   Resid.Df      D Df.diff  P.value
## 1      455  0.00000      0
## 2      442 15.75468     13 0.262631
```

NO3 does not improve the model.

Does NO3 improve the model?

```
AIC(model7, model8)
```

```
##           df      AIC
## model7  39 2663.824
## model8  26 2653.578
```

There are 14 more parameters in model7 than model8, so AIC needs to be 14×2 points lower.

Comparison to an “ordinary” NB GLM

```
model9 <- gllvm::gllvm(y, X = X, formula = ~N03, family = "negative.binomi
                      disp.formula = rep(1, ncol(y)), num.lv = 0)
```

```
data.frame(VGLM = c(coef(model7, "Xcoef")), GLM = c(coef(model9, "Xcoef")))
```

##	VGLM	GLM
## 1	0.02397401	0.02400093
## 2	-1.13607690	-1.14307572
## 3	0.24073469	0.24067658
## 4	0.43204257	0.43191554
## 5	0.21026169	0.21574188
## 6	0.11861631	0.11916282
## 7	-0.34156357	-0.36176377
## 8	0.05312437	0.05386581
## 9	0.45120511	0.45205829
## 10	-1.52495149	-1.36752654
## 11	0.59121431	0.59062702

Comparison NB GLM: standard errors

```
data.frame("VGLM" = c(model7$sd$Xcoef), "GLM" = c(model9$sd$Xcoef))
```

##	VGLM	GLM
## 1	0.2807960	0.3835160
## 2	0.3659817	0.4283106
## 3	0.5910521	0.4062606
## 4	0.3070574	0.3435355
## 5	0.4035969	0.5086282
## 6	0.2106619	0.3928671
## 7	0.5535621	0.4188497
## 8	0.3553290	0.4292095
## 9	0.3627138	0.4282178
## 10	0.9379137	0.5400633
## 11	0.4383276	0.4185578
## 12	0.6880651	0.3973890
## 13	0.4942584	0.4538790

Downsides of VGLMs

- ▶ One dispersion parameter per species: more parameters to estimate
- ▶ VGLM assumes 1 parameter per species per covariate
- ▶ This does not tend to work very well for real (sparse) community data
- ▶ VGLM assumes independence of species
- ▶ Does not include random effects (pseudoreplication, autocorrelation); VGLMMs address this next

Summary

- ▶ GLMs are fun, but not usually suitable for multispecies data
- ▶ VGLMs; fitting one model per species gives more flexibility
- ▶ This facilitates adding components that are shared across species
- ▶ Which is especially helpful when working with random effects

So far we have assumed that species do not influence each other